OPERATING SYSTEM

ORANGE PROBLEM-2

NAME:- DHAREEN.G

SRN:-PES1UG24AM084

2ND YEAR AIML

'B' SECTION

```
● @IAmKai1 ➜/workspaces/PES1UG24AM084-pes-vcs1 (main) $ make clean
  rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
  rm -rf .pes
● @IAmKai1 ➜/workspaces/PES1UG24AM084-pes-vcs1 (main) $ make all
  gcc -Wall -Wextra -O2 -c object.c -o object.o
  gcc -Wall -Wextra -O2 -c tree.c -o tree.o
  gcc -Wall -Wextra -O2 -c index.c -o index.o
  gcc -Wall -Wextra -O2 -c commit.c -o commit.o
  gcc -Wall -Wextra -O2 -c pes.c -o pes.o
  gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
  gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
  gcc -o test_objects test_objects.o object.o -lcrypto
  gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
  gcc -o test_tree test_tree.o object.o tree.o -lcrypto
● @IAmKai1 ➜/workspaces/PES1UG24AM084-pes-vcs1 (main) $ ./pes init
  echo "Hello" > hello.txt
  ./pes add hello.txt
  export PES_AUTHOR="Dhareen G <PES1UG24AM084>"
  ./pes commit -m "Initial commit"
  ./pes log
  Initialized empty PES repository in .pes/
  Committed: 4824bd474707... Initial commit
  commit 4824bd47470752df84e1ed567b95bd76115d65f0b64a8a769dfd0b33c8791dd1
  Author: Dhareen G <PES1UG24AM084>
  Date:   1776764154

      Initial commit
```

```
● @IAmKai1 ➜/workspaces/PES1UG24AM084-pes-vcs1 (main) $ ./pes log
find .pes -type f | sort
cat .pes/refs/heads/main
cat .pes/HEAD
commit 4824bd47470752df84e1ed567b95bd76115d65f0b64a8a769dfd0b33c8791dd1
Author: Dhareen G <PES1UG24AM084>
Date:   1776764154

    Initial commit

.pes/HEAD
.pes/index
.pes/objects/48/24bd47470752df84e1ed567b95bd76115d65f0b64a8a769dfd0b33c8791dd1
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/refs/heads/main
4824bd47470752df84e1ed567b95bd76115d65f0b64a8a769dfd0b33c8791dd1
ref: refs/heads/main
```

```
● @IAmKai1 ➜/workspaces/PES1UG24AM084-pes-vcs1 (main) $ echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"

./pes log
Committed: 45579aa991ea... Add world
Committed: 4e446eb6f879... Add farewell
commit 4e446eb6f879ce6b9baf68c4d5347244eee05f9eca269f97e406099a04a55c63
Author: Dhareen G <PES1UG24AM084>
Date:   1776764177

    Add farewell

commit 45579aa991ea4eeb3de2329157e0c25256e4c421a168f81246d3a81be063f544
Author: Dhareen G <PES1UG24AM084>
Date:   1776764177

    Add worldell

commit 4824bd47470752df84e1ed567b95bd76115d65f0b64a8a769dfd0b33c8791dd1
Author: Dhareen G <PES1UG24AM084>
Date:   1776764154
```

```
@IAmKai1 ➜/workspaces/PES1UG24AM084-pes-vcs1 (main) $ echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt …
Author: Dhareen G <PES1UG24AM084>
Date:   1776764154

    Initial committial comQ
```

```
@IAmKai1 ➜/workspaces/PES1UG24AM084-pes-vcs1 (main) $ make test_objects
./test_objects

make test_tree
./test_tree
make: 'test_objects' is up to date.
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
make: 'test_tree' is up to date.
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

```
● @IAmKai1 ➜/workspaces/PES1UG24AM084-pes-vcs1 (main) $ make test-integration
  === Running integration tests ===
  bash test_sequence.sh
  === PES-VCS Integration Test ===

  --- Repository Initialization ---
  Initialized empty PES repository in .pes/
  PASS: .pes/objects exists
  PASS: .pes/refs/heads exists
  PASS: .pes/HEAD exists

  --- Staging Files ---
  Status after add:
  Staged changes:
    staged:     file.txt
    staged:     hello.txt

  Unstaged changes:
    (nothing to show)

  Untracked files:
    (nothing to show)


  --- First Commit ---
  Committed: e68c93bcc2db... Initial commit
```

```
@IAmKai1 ➜/workspaces/PES1UG24AM084-pes-vcs1 (main) $ make test-integration

Log after first commit:
commit e68c93bcc2db5aa2f50cc9ae863f400b9d0817d0897845592f70e32a38fb3c5a
Author: Dhareen G <PES1UG24AM084>
Date:   1776764198

    Initial commit


--- Second Commit ---
Committed: 42594c1054cc... Update file.txt

--- Third Commit ---
Committed: 56b7eabcdff4... Add farewell

--- Full History ---
commit 56b7eabcdff48f0ffd582af8b8e40a3b5dfd9cc5383bb6dbbda7732859db8bd6
Author: Dhareen G <PES1UG24AM084>
Date:   1776764198

    Add farewell

commit 42594c1054cc4f9cd276f92ba55b8c8de3e8a0660193722652b57684d50d5ad1
Author: Dhareen G <PES1UG24AM084>
Date:   1776764198

    Update file.txt

commit e68c93bcc2db5aa2f50cc9ae863f400b9d0817d0897845592f70e32a38fb3c5a
Author: Dhareen G <PES1UG24AM084>
```

```
@IAmKai1 ➜/workspaces/PES1UG24AM084-pes-vcs1 (main) $ ./test_objects
./test_tree
./pes log
find .pes -type f | sort
cat .pes/refs/heads/main …
    Update file.txt

commit 0f589ebe7669bfafde05672fb62bfc5356b8e8d4bacbd9450f71c5628992617f
Author: Dhareen G <PES1UG24AM084>
Date:   1776764257

    Initial committial comQ


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
d78d21a35213b53f270c4c8a5767058ba0c62653b69a99181d0127b0c6f01979

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/0f/589ebe7669bfafde05672fb62bfc5356b8e8d4bacbd9450f71c5628992617f
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/3b/41bbff4aae154cfe52cda46b15f59213b9953c8f3be8f8413cae2d39acccd2
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
```